MACH1 / FleetLine E1 — The argument, in six steps

1 · The Problem

Indian ride-hailing fleets run on CNG — not by preference, because nothing was built for their duty cycle. Fleet duty is 200 km/day, 312 days a year, return-to-base every night. It is the most demanding duty cycle an EV can have, and the one the Indian market has served worst.

2 · The Insight

Volume and margin live in different segments, and they need different vehicles.

VOLUME — not the launch segment

- street-hail intra-city · ~58% of rides
- unpredictable dispatch · price sensitivity 0.85 · ~10% EV adoption

MARGIN — the launch segment

- airport + corporate commute · ~19% combined
- 144–210 paid km/day · pre-booked
- price sensitivity 0.30–0.35 · ~18% EV adoption

We did not chase the largest segment. We launched where the economics close and EV adoption already exists.

3 · The Vehicle

BEV · LFP 45 kWh · PMSM 80 bhp · **268 km at 15% reserve** (315 km at zero) · 3,995 mm — deliberately 5 mm inside the small-car tax line.

Designed around one number: 200 km/day, return-to-base, one 7.4 kW AC port, overnight.

4 · The Math

NPV at a 12% hurdle, escalated, **with the ₹99,000/vehicle depot charger inside the cost boundary**.

	Delhi	Mumbai	Bengaluru
5-year	**+6.7%**	+4.0%	+0.9%
8-year	**+14.6%**	+10.6%	+9.0%

All three clear at five years. **One cell in the whole 48-cell model is negative** — Bengaluru, bought cash, −0.4%.
Broker and platform fees are identical on either powertrain, so the EV-vs-CNG difference is unaffected by how commission is structured.

Inter-city: Delhi–Agra 200, Mumbai–Pune 150, Mumbai–Nashik 170, Bengaluru–Mysuru 145 all fit inside the 268 km envelope. Delhi–Jaipur 280 is marginal. Beyond ~300 km, destination charging is a fleet-contract requirement, not a vehicle spec.

5 · Why LFP, and what we admit

Published LFP studies disagree **3.6×** on time-to-80%. So per-cell State-of-Health telemetry is specified, **not optional** — it is the mechanism that makes an 8-year capacity-retention guarantee credible.

6 · The Risk We Cannot Hedge

No central scheme pays for an electric passenger car. PM E-DRIVE's ₹11,900 cr covers 2W, 3W, rickshaws, ambulances, trucks, buses and charging — not 4W cars. Maharashtra does pay: **15% of ex-factory, capped ₹2 lakh, for transport e-4W, 25,000 units.**

And the concessions we do rely on — Delhi's life-of-vehicle road-tax exemption, Maharashtra's e-4W incentive — **both expire 31 March 2030**, five years before the target year. Renewing them is a policy position worth about five points, not an engineering one.

1,000–2,000 vehicles, Delhi, gated on cell supply · measured duty · twelve months of our own battery data.